Discussion section 2/4/16 summary

$$\vec{A}(\vec{r},t) = \frac{\mu_0}{4\pi r}\int d^3r\, \underbrace{j\left(\vec{r}\,', t - \frac{r}{c} + \frac{\vec{r}\,'\cdot\vec{n}}{c}\right)}$$

Fourier transform $\longrightarrow \displaystyle\int_{-\infty}^{\infty} \frac{d\omega}{2\pi}\, j(\vec{r}\,',\omega)\, e^{-i\omega\left(t - \frac{r}{c} + \frac{\vec{r}\,'\cdot\vec{n}}{c}\right)}$

$$= \frac{\mu_0}{4\pi r}\int_{-\infty}^{\infty}\frac{d\omega}{2\pi}\, e^{-i\omega(t-\frac{r}{c})}\int d^3r'\, j(\vec{r}\,',\omega)\, e^{-i\frac{\omega}{c}\vec{r}\,'\cdot\vec{n}}$$

$\vec{k} = \frac{\omega}{c}\vec{n}$

$$= \frac{\mu_0}{4\pi r}\int_{-\infty}^{\infty}\frac{d\omega}{2\pi}\, e^{-i\omega(t-\frac{r}{c})}\, \underbrace{j\left(\frac{\omega}{c}\vec{n},\omega\right)}$$

$\longrightarrow$ zeroth order $\vec{k} = 0$ gives dipole radiation

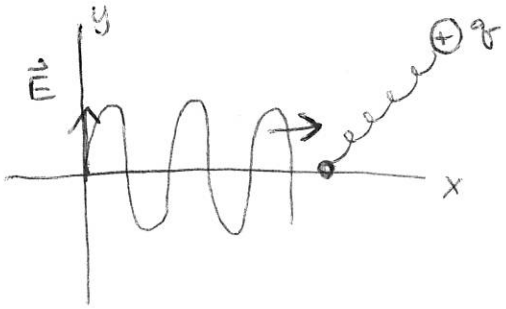
expand $\rightarrow$ Legendre polynomials

$$\vec{j}_\omega(\vec{n}) = \sum_{lm} Y_{lm}(\vec{n})\, j_{lm}$$

Example problem: a point charge $(+q)$ is connected to a nucleus by a spring with $k = m\omega_0^2$.

① write equation of motion for $\vec{r}$
② solve for $\vec{r}(t)$
③ how much energy is radiated?

} consider two extremes, $k \approx 0$, $k \approx \infty$, consider radiative friction



$$\vec{E} = \vec{E}_0 e^{-i\omega t}$$

$$\vec{F}_{rf} = \frac{\mu_0 q \dddot{\vec{d}}}{6\pi c} = \overbrace{\frac{\mu_0 q^2 \dddot{\vec{r}}}{6\pi c}}^{\text{for a single point charge}}$$

$$\vec{F}_{sp} = -m\omega_0^2\vec{r}\,, \quad \vec{F}_e = q\vec{E}$$

$\therefore$ equation of motion:

① $$\boxed{m\ddot{\vec{r}} = -m\omega_0^2\vec{r} + q\vec{E}_0 e^{-i\omega t} + \frac{\mu_0 q^2 \dddot{\vec{r}}}{6\pi c}}$$

② try $\vec{r} = \vec{r}_0 e^{-i\omega t}$

$$-m\omega^2 \vec{r}_0 = -m\omega_0^2 \vec{r}_0 + q\vec{E}_0 + \frac{i\omega^3 \mu_0 q^2 \vec{r}_0}{6\pi c}$$

$$\vec{r}_0\left(m\omega_0^2 - m\omega^2 - \frac{i\omega^3\mu_0 q^2}{6\pi c}\right) = q\vec{E}_0$$

$$\vec{r}_0 = \underbrace{q\vec{E}_0 \Big/ \left(m\omega_0^2 - m\omega^2 - \frac{i\omega^3\mu_0 q^2}{6\pi c}\right)}_{\boxed{\vec{r} = \text{"}\downarrow\text{"}\, e^{-i\omega t}}} \Bigg\} \text{ inhomogenous solution}$$

③ $$\frac{dP}{d\Omega} = \frac{\mu_0}{16\pi^2 c}\left(\ddot{\vec{d}} \times \hat{n}\right) = \frac{\mu_0 q^2 \ddot{\vec{r}}^{\,2}}{16\pi^2 c}\sin^2\theta$$

$$\vec{d} = q\vec{r}(t) = \frac{q^2 \vec{E}_0}{\boxed{m(\omega_0^2 - \omega^2) - \frac{i\omega^3\mu_0 q^2}{6\pi c}}} e^{-i\omega t}$$

linearly oscillating dipole example from book pg. 356

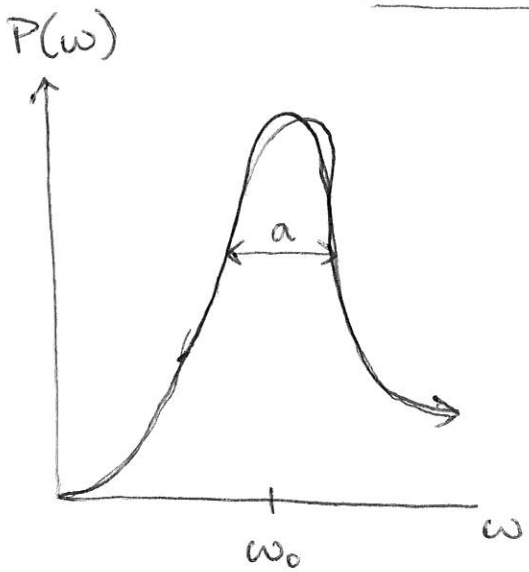
boxed quantity = d_0

eq. 12.52: $$\frac{dP}{d\Omega} = \frac{\mu_0 \omega^4 d_0^2}{32\pi^2 c}\sin^2\theta$$

$$P = \int \frac{dP}{d\Omega}\, d\Omega = \frac{\mu_0 \omega^4 |\vec{d}_0|^2}{12\pi c}$$

this result is $\frac{1}{2}$ result given in the book because time average has already been taken.

Power dependence on frequency



① $\omega \ll \omega_0$

② $\omega \sim \omega_0$

③ $\omega \gg \omega_0$

$$|\vec{d}_0| = \frac{q^2 E_0}{\sqrt{\left[m(\omega_0^2-\omega^2)\right]^2 + \left(\frac{\omega^3 \mu_0 q^2}{6\pi c}\right)^2}}$$

$$P = \frac{\mu_0 \omega^4}{12\pi c}\left[\left[m(\omega_0^2-\omega^2)\right]^2 + \left(\frac{\omega^3 \mu_0 q^2}{6\pi c}\right)^2\right]^{-1}$$

① P approaches zero.

② $$\approx \frac{\omega_0^2}{\alpha(\omega_0+\omega)^2\underbrace{(\omega_0-\omega)^2}_{\text{can't set this difference to zero because it defines shape of peak.}} + \beta\omega_0^4}$$

} damping term prevents this from going to infinity

$$a = \text{width} \sim \left(\omega_0^2 \times \frac{\mu_0 q^2}{6m\pi c}\right) \longleftarrow \omega_0 \times \underbrace{\frac{\omega_0 \mu_0 q^2}{6m\pi c}}_{\text{this should be dimensionless}}$$

③ high frequency limit, defined by radiation friction term

"Thomson Scattering"